

# Finger Pose Classification — Run Report

Generated 2026-05-19 15:09:39

## Data paths

|                            |                                                                              |
|----------------------------|------------------------------------------------------------------------------|
| Data root                  | /home/jestin/ThesisRepo/ML/NewTrainingData/Finger_MCP_PIP · 121 trials total |
| Class folders              | curl_curl, straight_straight                                                 |
| Report output dir          | /home/jestin/ThesisRepo/ML/NewReports/FingerPoseReports/Binary               |
| Trials · curl_curl         | 61                                                                           |
| Trials · straight_straight | 60                                                                           |

## Sensor selection

|                |                                             |
|----------------|---------------------------------------------|
| Hands          | left, right                                 |
| Segments       | palm_mid, thumb, index, middle, ring, pinky |
| Modalities     | YPR, Accel, Flex                            |
| Sensor columns | 164 channels                                |
| Fingers        | thumb, index, middle, ring, pinky           |
| Classes        | 2: curl_curl, straight_straight             |

## Preprocessing, features & split

|                           |                                              |
|---------------------------|----------------------------------------------|
| Resample steps            | 90                                           |
| Butterworth               | on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz   |
| Normalisation             | standard                                     |
| Feature mode              | stats+fft → feature dim 1080                 |
| Split / seed / CV         | test_size=0.2 · random_state=42 · CV folds=5 |
| Trials per finger dataset | 409 (train+test, post-aug train)             |

## Augmentation (training only)

|                        |                                          |
|------------------------|------------------------------------------|
| Copies per train trial | 3                                        |
| Gaussian noise         | on · sigma=0.02 (per-channel std-scaled) |
| Time warp              | on · max_scale=0.1                       |
| Amplitude scaling      | off · range=(0.9, 1.1)                   |

## Active classifiers

| Algorithm                       | Class                      |
|---------------------------------|----------------------------|
| SVM (RBF)                       | SVC                        |
| SVM (Linear)                    | SVC                        |
| Random Forest                   | RandomForestClassifier     |
| KNN                             | KNeighborsClassifier       |
| Logistic Regression             | LogisticRegression         |
| LDA                             | LinearDiscriminantAnalysis |
| Voting Soft (SVM+RF+KNN+LogReg) | VotingClassifier           |
| Voting Soft (SVM+RF)            | VotingClassifier           |
| Stacking (SVM+RF+KNN+LogReg)    | StackingClassifier         |

## Classifier hyperparameters

|                                        |                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SVM (RBF)</b>                       | C=10 · break_ties=False · cache_size=200 · class_weight=None · coef0=0.0 · decision_function_shape=ovr · degree=3 · gamma=scale · kernel=rbf · max_iter=-1 · probability=True · random_state=42 · shrinking=True · tol=0.001 · verbose=False                                                                                                                |
| <b>SVM (Linear)</b>                    | C=1 · break_ties=False · cache_size=200 · class_weight=None · coef0=0.0 · decision_function_shape=ovr · degree=3 · gamma=scale · kernel=linear · max_iter=-1 · probability=True · random_state=42 · shrinking=True · tol=0.001 · verbose=False                                                                                                              |
| <b>Random Forest</b>                   | bootstrap=True · ccp_alpha=0.0 · class_weight=None · criterion=gini · max_depth=None · max_features=sqrt · max_leaf_nodes=None · max_samples=None · min_impurity_decrease=0.0 · min_samples_leaf=1 · min_samples_split=2 · min_weight_fraction_leaf=0.0 · monotonic_cst=None · n_jobs=-1 · oob_score=False · random_state=42 · verbose=0 · warm_start=False |
| <b>KNN</b>                             | algorithm=auto · leaf_size=30 · metric=euclidean · metric_params=None · n_jobs=None · n_neighbors=5 · p=2 · weights=uniform                                                                                                                                                                                                                                 |
| <b>Logistic Regression</b>             | C=1.0 · class_weight=None · dual=False · fit_intercept=True · intercept_scaling=1 · l1_ratio=0.0 · max_iter=1000 · n_jobs=None · penalty=deprecated · random_state=42 · solver=lbfgs · tol=0.0001 · verbose=0 · warm_start=False                                                                                                                            |
| <b>LDA</b>                             | covariance_estimator=None · n_components=None · priors=None · shrinkage=None · solver=svd · store_covariance=False · tol=0.0001                                                                                                                                                                                                                             |
| <b>Voting Soft (SVM+RF+KNN+LogReg)</b> | flatten_transform=True · n_jobs=-1 · verbose=False · voting=soft · weights=None                                                                                                                                                                                                                                                                             |
| <b>Voting Soft (SVM+RF)</b>            | flatten_transform=True · n_jobs=-1 · verbose=False · voting=soft · weights=None                                                                                                                                                                                                                                                                             |
| <b>Stacking (SVM+RF+KNN+LogReg)</b>    | cv=5 · final_estimator=LogisticRegression(max_iter=1000, random_state=42) · n_jobs=-1 · passthrough=False · stack_method=auto · verbose=0                                                                                                                                                                                                                   |

## Per-dataset test accuracy

Rows = (hand, finger) dataset. Columns = classifier. Last row = mean across datasets.

| Dataset        | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SVM+RF+KNN+LogReg) | Voting Soft (SVM+RF) | Stacking (SVM+RF+KNN+LogReg) |
|----------------|-----------|--------------|---------------|-------|---------------------|-------|---------------------------------|----------------------|------------------------------|
| left · thumb   | 0.840     | 0.920        | 0.920         | 0.880 | 0.920               | 0.800 | 1.000                           | 1.000                | 0.960                        |
| left · index   | 0.960     | 0.960        | 1.000         | 0.880 | 0.960               | 0.880 | 0.960                           | 0.960                | 0.960                        |
| left · middle  | 0.960     | 0.920        | 0.960         | 0.800 | 0.920               | 0.880 | 0.920                           | 0.920                | 0.920                        |
| left · ring    | 0.920     | 0.960        | 0.920         | 0.880 | 0.960               | 0.920 | 0.960                           | 0.960                | 0.960                        |
| left · pinky   | 0.880     | 0.800        | 0.960         | 0.960 | 0.840               | 0.760 | 0.960                           | 0.960                | 0.880                        |
| right · thumb  | 1.000     | 0.960        | 0.960         | 0.960 | 0.960               | 0.960 | 1.000                           | 1.000                | 1.000                        |
| right · index  | 0.960     | 0.960        | 0.960         | 0.920 | 0.960               | 0.960 | 0.960                           | 0.960                | 0.960                        |
| right · middle | 0.920     | 0.920        | 0.960         | 0.920 | 0.920               | 0.880 | 0.920                           | 0.920                | 0.920                        |
| right · ring   | 0.960     | 0.960        | 0.960         | 0.960 | 0.960               | 0.880 | 0.960                           | 0.960                | 0.960                        |
| right · pinky  | 0.920     | 0.880        | 0.960         | 0.960 | 0.880               | 0.920 | 0.920                           | 0.920                | 0.880                        |
| Mean           | 0.932     | 0.924        | 0.956         | 0.912 | 0.928               | 0.884 | 0.956                           | 0.956                | 0.940                        |

## Per-class accuracy by dataset

Each table: rows = class, columns = classifier. Values are per-class accuracy on the test set.

### left · thumb

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SVM+RF+KNN+LogReg) | Voting Soft (SVM+RF) | Stacking (SVM+RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|---------------------------------|----------------------|------------------------------|
| curl_curl         | 0.692     | 0.846        | 0.923         | 0.923 | 0.846               | 0.769 | 1.000                           | 1.000                | 0.923                        |
| straight_straight | 1.000     | 1.000        | 0.917         | 0.833 | 1.000               | 0.833 | 1.000                           | 1.000                | 1.000                        |

### left · index

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SVM+RF+KNN+LogReg) | Voting Soft (SVM+RF) | Stacking (SVM+RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|---------------------------------|----------------------|------------------------------|
| curl_curl         | 1.000     | 1.000        | 1.000         | 0.923 | 1.000               | 0.846 | 1.000                           | 1.000                | 1.000                        |
| straight_straight | 0.917     | 0.917        | 1.000         | 0.833 | 0.917               | 0.917 | 0.917                           | 0.917                | 0.917                        |

### left · middle

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SVM+RF+KNN+LogReg) | Voting Soft (SVM+RF) | Stacking (SVM+RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|---------------------------------|----------------------|------------------------------|
| curl_curl         | 1.000     | 0.923        | 1.000         | 0.923 | 0.923               | 0.846 | 0.923                           | 0.923                | 0.923                        |
| straight_straight | 0.917     | 0.917        | 0.917         | 0.667 | 0.917               | 0.917 | 0.917                           | 0.917                | 0.917                        |

### left · ring

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SV M+RF+K NN+Log Reg) | Voting Soft (SV M+RF) | Stacking (SVM+ RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|------------------------------------|-----------------------|-------------------------------|
| curl_curl         | 0.923     | 1.000        | 1.000         | 0.923 | 1.000               | 0.846 | 0.923                              | 0.923                 | 0.923                         |
| straight_straight | 0.917     | 0.917        | 0.833         | 0.833 | 0.917               | 1.000 | 1.000                              | 1.000                 | 1.000                         |

#### left · pinky

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SV M+RF+K NN+Log Reg) | Voting Soft (SV M+RF) | Stacking (SVM+ RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|------------------------------------|-----------------------|-------------------------------|
| curl_curl         | 0.769     | 0.615        | 0.923         | 0.923 | 0.692               | 0.615 | 0.923                              | 0.923                 | 0.769                         |
| straight_straight | 1.000     | 1.000        | 1.000         | 1.000 | 1.000               | 0.917 | 1.000                              | 1.000                 | 1.000                         |

#### right · thumb

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SV M+RF+K NN+Log Reg) | Voting Soft (SV M+RF) | Stacking (SVM+ RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|------------------------------------|-----------------------|-------------------------------|
| curl_curl         | 1.000     | 0.923        | 0.923         | 1.000 | 0.923               | 0.923 | 1.000                              | 1.000                 | 1.000                         |
| straight_straight | 1.000     | 1.000        | 1.000         | 0.917 | 1.000               | 1.000 | 1.000                              | 1.000                 | 1.000                         |

#### right · index

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SV M+RF+K NN+Log Reg) | Voting Soft (SV M+RF) | Stacking (SVM+ RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|------------------------------------|-----------------------|-------------------------------|
| curl_curl         | 0.923     | 0.923        | 0.923         | 0.923 | 0.923               | 0.923 | 0.923                              | 0.923                 | 0.923                         |
| straight_straight | 1.000     | 1.000        | 1.000         | 0.917 | 1.000               | 1.000 | 1.000                              | 1.000                 | 1.000                         |

#### right · middle

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SV M+RF+K NN+Log Reg) | Voting Soft (SV M+RF) | Stacking (SVM+ RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|------------------------------------|-----------------------|-------------------------------|
| curl_curl         | 0.923     | 0.923        | 0.923         | 0.923 | 0.923               | 0.923 | 0.923                              | 0.923                 | 0.923                         |
| straight_straight | 0.917     | 0.917        | 1.000         | 0.917 | 0.917               | 0.833 | 0.917                              | 0.917                 | 0.917                         |

#### right · ring

| Finger            | SVM (RBF) | SVM (Linear) | Random Forest | KNN   | Logistic Regression | LDA   | Voting Soft (SV M+RF+K NN+Log Reg) | Voting Soft (SV M+RF) | Stacking (SVM+ RF+KNN+LogReg) |
|-------------------|-----------|--------------|---------------|-------|---------------------|-------|------------------------------------|-----------------------|-------------------------------|
| curl_curl         | 0.923     | 0.923        | 0.923         | 0.923 | 0.923               | 0.846 | 0.923                              | 0.923                 | 0.923                         |
| straight_straight | 1.000     | 1.000        | 1.000         | 1.000 | 1.000               | 0.917 | 1.000                              | 1.000                 | 1.000                         |

#### right · pinky

| Finger            | SVM<br>(RBF) | SVM<br>(Linear) | Random<br>Forest | KNN   | Logistic<br>Regress<br>ion | LDA   | Voting<br>Soft (SV<br>M+RF+K<br>NN+Log<br>Reg) | Voting<br>Soft (SV<br>M+RF) | Stackin<br>g (SVM+<br>RF+KN<br>N+LogR<br>eg) |
|-------------------|--------------|-----------------|------------------|-------|----------------------------|-------|------------------------------------------------|-----------------------------|----------------------------------------------|
| curl_curl         | 0.923        | 0.923           | 0.923            | 1.000 | 0.923                      | 0.846 | 0.923                                          | 0.923                       | 0.923                                        |
| straight_straight | 0.917        | 0.833           | 1.000            | 0.917 | 0.833                      | 1.000 | 0.917                                          | 0.917                       | 0.833                                        |